NCCP 00222 · VERSION 5

Your treatment, explained

Pemetrexed Monotherapy

21-day cycle

Source reconciled; independent clinical, pharmacy and patient/PPI review pending

ACT ON SYMPTOMS, NOT THE CALENDAR

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Call your oncology 24-hour number immediately for a temperature of 37.5°C or above, 36°C or below, shivering or sudden illness. Call urgently for new breathlessness, chest pain, bleeding, severe or persistent diarrhoea, vomiting despite medicines, inability to drink, very little urine or a spreading or blistering rash. Do not wait for the next visit; serious infection can occur without a high temperature.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

Open your CUH contact / treatment card

Sources: hse · smpc

Choose a topic to read

YOUR COMPLETE REFERENCE

Read what you need, when you need it.

Start with the overview, explore the cycle, or go straight to a question. Urgent advice is always available.

- How each drug works
- A three-week cycle within an ongoing treatment plan
- Your 21-day timeline
- Baseline and cycle-specific checks
- Whole treatment, scans and checks
- Tablets, visits and practical planning
- Understanding your treatment
- Before starting or changing to maintenance
- Treatment day
- Care between visits
- When to get urgent help
- Review and the next plan
- Vaccines and radiotherapy
- Supportive medicines
- Understanding the evidence
- Questions to ask

Sources

UNDERSTAND YOUR TREATMENT

Your chemotherapy medicine

Pemetrexed is chemotherapy used to control selected non-squamous non-small-cell lung cancer that is locally advanced or has spread. Maintenance aims to keep cancer controlled after platinum chemotherapy has finished without the cancer progressing. Second-line treatment is a separate treatment decision after previous chemotherapy. Ask which aim and indication apply to you.

ANTIFOLATE CHEMOTHERAPY

Pemetrexed

Alimta



Interrupts DNA building-block production

Pemetrexed blocks several enzymes that use folate to make the building blocks of DNA and RNA. This can slow or stop cancer-cell growth. Healthy cells can also be affected. Prescribed folic acid and vitamin B12 reduce important toxicities and are part of treatment.

Folate-dependent enzymes blocked → DNA building blocks reduced → Cell growth disrupted

What to watch for

Take the prescribed folic acid and attend for B12. Report mouth sores, diarrhoea, a rash, breathlessness or reduced urine; check before taking ibuprofen or other non-prescription medicines.

Sources: nccp · smpc

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Explore your existing anatomy and side-effect map

THE BIGGER TREATMENT PLAN

A three-week cycle within an ongoing treatment plan

DAY 1 INFUSION EVERY THREE WEEKS

Pemetrexed alone: treatment, recovery and review

D1 D2 D3–D20 D21

Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

Sources: nccp

BASELINE → TREATMENT → FOLLOW-UP

Checks through your pemetrexed treatment

These columns show the order of care, not equal time intervals. Baseline tests and tests before every cycle are distinct. There is no fixed treatment length or scan interval; any additional immunotherapy needs its own monitoring plan.

Check	Before starting / phase change	Each 21-day cycle	Between visits	Benefit and toxicity review	After last pemetrexed
Blood count, kidneys and liver	Baseline FBC, renal and liver profile	Repeat before each cycle	Earlier tests if clinically needed	Review recovery and any dose change	Follow unresolved abnormalities
Vitamins and steroids	Confirm folate, last B12 and one steroid schedule	Folate daily; B12 every three cycles; steroid around each dose	Report missed medicines or poor intake	Check adherence and changes to support medicines	Continue folate for 21 days after final dose

Check	Before starting / phase change	Each 21-day cycle	Between visits	Benefit and toxicity review	After last pemetrexed
Symptoms and medicine interactions	Review prior toxicity and all medicines	Assess interval symptoms before authorising treatment	Urgent contact for illness; check NSAID/ aspirin use	Discuss prior reductions, nerve recurrence and daily function	Follow persistent kidney, nerve, blood or other problems
Disease and treatment aim	Confirm maintenance versus second-line indication	Review symptoms and clinical progress	Extra assessment if needed	Scans and tests according to Consultant plan	Agree follow-up or a different treatment plan

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00222 v5 · official regimen · EMA ALIMTA product information and patient leaflet

THE WHOLE TREATMENT JOURNEY

Pemetrexed Monotherapy: treatment and checks

Non-squamous NSCLC · standalone maintenance or second-line chemotherapy

One pemetrexed infusion on Day 1 of each 21-day cycle. Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

On a small screen, swipe the map sideways. The checks below give the same timing in words.

Treatment, tests and reviews across the course

Care / check	Before starting / phase change	Each 21-day cycle	Between visits	Benefit and toxicity review	After last pemetrexed
Treatment course NCCP 00222	Confirm standalone maintenance or second-line	Day 1 pemetrexed; repeat every 21 days if appropriate	Recovery, daily folate and symptom reporting	Continue only while benefit and tolerability remain acceptable	Agree next treatment or follow-up plan
Blood count, kidneys and liver NCCP 00222	Baseline FBC, renal and liver profile	Repeat before each cycle	Earlier tests if clinically needed	Review recovery and any dose change	Follow unresolved abnormalities
Vitamins and steroids NCCP 00222	Confirm folate, last B12 and one steroid schedule	Folate daily; B12 every three cycles; steroid around each dose	Report missed medicines or poor intake	Check adherence and changes to support medicines	Continue folate for 21 days after final dose
Symptoms and medicine interactions NCCP 00222 · SMPC	Review prior toxicity and all medicines	Assess interval symptoms before authorising treatment	Urgent contact for illness; check NSAID/ aspirin use	Discuss prior reductions, nerve recurrence and daily function	Follow persistent kidney, nerve, blood or other problems

Care / check	Before starting / phase change	Each 21-day cycle	Between visits	Benefit and toxicity review	After last pemetrexed
Disease and treatment aim NCCP 00222	Confirm maintenance versus second-line indication	Review symptoms and clinical progress	Extra assessment if needed	Scans and tests according to Consultant plan	Agree follow-up or a different treatment plan

What each check is for

1

Vitamin preparation or carry-forward

Before first pemetrexed or a phase change

Vitamins lower serious pemetrexed toxicity.

Ask / act: Confirm last B12 and next due date, daily folate and one steroid prescription.

Sources: NCCP 00222

2

Review each dose

Before every 21-day cycle

Counts, organ function and toxicity can change.

Ask / act: Confirm whether to continue, delay, modify or stop.

Sources: NCCP 00222

3

Benefit review

According to Consultant plan

There is no fixed total cycle number.

Ask / act: Confirm ongoing benefit and the next plan.

Sources: NCCP 00222

4

Final dose follow-up

After the final pemetrexed

Protection and recovery continue after the infusion.

Ask / act: Agree medicine stop dates and review persistent symptoms.

Sources: NCCP 00222

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Sources checked 2026-11-02 · Source reconciled; independent clinical, pharmacy and patient/PPI review pending · SACTCheck v0.84.29

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

One pemetrexed infusion on Day 1 of each 21-day cycle. Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.



Before the first pemetrexed: Folate preparation and the first B12 injection start before the infusion. When already on pemetrexed, confirm the established vitamin schedule.

Treatment length: Treatment is planned every three weeks while it is helping and side effects

Day before each dose: Start the prescribed rash-prevention steroid the day before pemetrexed; the reference cycle starts at infusion Day 1.

remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

After the final dose: Folate continues for 21 days after the last pemetrexed; persistent symptoms and follow-up remain important.

> **Why these bands stay broad:** This reference cycle shows care and checks, not predicted safe or high-risk days. Symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00222 v5 · official regimen ↗

> **There are no guaranteed “safe days”.** Follow your unit’s emergency instructions and use its 24-hour number. Serious infection can occur without a high temperature; sudden illness or shivering still needs immediate advice.

Read the whole cycle as text

Day 1: Infusion day

Pemetrexed is infused into a vein over about 10 minutes on Day 1. The whole visit is longer because it includes blood-result review, symptom checks, medicines, preparation and observation. The team can use a cannula or suitable existing line; this regimen has no routine take-home infusion pump. Report any reaction or discomfort around the drip immediately.

Day 2: Day after pemetrexed

Continue the prescribed rash-prevention steroid through the day after treatment, folic acid daily, and anti-sickness medicine as directed. Use one reconciled medicine plan; do not double steroids.

Day 3–20: Between visits

Continue folic acid and monitor symptoms. Your B12 injection is due every three cycles and may be booked on a treatment day. Call promptly for illness, poor intake, reduced urine, bowel or mouth symptoms; do not wait for review.

Day 21: Review before another cycle

Blood count, kidney and liver tests, symptoms, previous toxicity and treatment benefit guide the next decision. Confirm the next dose, vitamins, support medicines and scan/review plan.

Practical treatment planning

Agree the arrangements with your oncology team.



STEP 1

Before the visit

Confirm blood-test arrangements, transport, prescriptions, B12 booking and the 24-hour number. Tell the team if you are ill before travelling.



STEP 2

Time in the unit

Pemetrexed is infused into a vein over about 10 minutes on Day 1. The whole visit is longer because it includes blood-result review, symptom checks, medicines, preparation and observation. The team can use a cannula or suitable existing line; this regimen has no routine take-home infusion pump.



STEP 3

Home medicines

Keep one written plan for folic acid, B12, steroid and anti-sickness medicines. Ask the pharmacist to resolve duplicate or conflicting instructions.



STEP 4

At a phase change

Confirm that the prescription is pemetrexed alone, review the anti-sickness plan, carry forward the appropriate vitamin schedule and ask whether other treatment needs separate information.



STEP 5

Practical support

Arrange help with travel or daily tasks and bring a medicines list, symptom notes and questions. Extra passport pages can be provided while treatment continues.

Open practical treatment guide · Printable practical guide

Sources: NCCP 00222 v5 · official regimen. Source reconciled; independent clinical, pharmacy and patient/PPI review pending.

MAKE THE PLAN PRACTICAL

Support around the medicines

1 Folic acid every day

The source uses folic acid 350–1000 micrograms daily. Take at least five doses in the seven days before the first pemetrexed, continue throughout treatment and for 21 days after the final dose. Follow your own prescription and tell the team about missed doses.

2 Vitamin B12 injection

The source uses vitamin B12 1000 micrograms into a muscle during the week before the first pemetrexed, then every three cycles. Later injections may be on treatment day. A change from platinum treatment to maintenance does not automatically restart the B12 clock: your team checks the last dose and the next due date.

3 One steroid prescription

Pemetrexed rash prevention uses dexamethasone-equivalent 4 mg twice daily on the day before, day of and day after treatment. Your pharmacist or prescriber should give one clear steroid schedule that also accounts for any anti-sickness medicines. Do not add a second steroid course yourself.

4 Low-risk anti-sickness plan

Pemetrexed on its own has a low sickness risk. Take the prescribed prevention and rescue medicines and call if sickness persists. A previous platinum anti-sickness course should be reviewed when moving to pemetrexed alone.

5 Kidneys and other medicines

Check with the team before using ibuprofen, other anti-inflammatory painkillers, aspirin or any new medicine. Kidney function and the medicine determine whether and when it needs interruption. Do not stop a prescribed medicine on your own; ask for a written plan.

6 Reporting symptoms and getting help

Keep your 24-hour oncology number available, record symptoms and ask for practical or emotional support early.

7 Vaccines

Ask the oncology team before any vaccination. Some live vaccines are unsuitable during treatment; the team can advise on the right vaccine and timing.

Sources: nccp · smpc · antiemetic

TREATMENT REFERENCE · 01

Understanding your treatment

Confirm the treatment aim and the medicines on your prescription.

Maintenance or second-line

Pemetrexed is chemotherapy used to control selected non-squamous non-small-cell lung cancer that is locally advanced or has spread. Maintenance aims to keep cancer controlled after platinum chemotherapy has finished without the cancer progressing. Second-line treatment is a separate treatment decision after previous chemotherapy. Ask which aim and indication apply to you.

Sources: nccp

One medicine, its own plan

This guide covers pemetrexed on its own for maintenance or second-line treatment of selected non-squamous non-small-cell lung cancer. If pembrolizumab or another immunotherapy is also prescribed, use the matching combination regimen and its additional consent, monitoring and urgent advice. Pemetrexed with pembrolizumab maintenance is a different treatment plan.

Sources: nccp

Treatment length

Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

Sources: nccp

TREATMENT REFERENCE · 02

Before starting or changing to maintenance

A change in phase needs a clear new prescription and review.

Checks before the dose

Blood count, kidney and liver tests are checked before starting and before each cycle. Tell the team about infections, breathlessness, bowel or mouth problems, rash, numbness and any previous severe treatment reaction.

Sources: nccp

Confirm the vitamin schedule

Check how much folic acid you take and when your last B12 injection was given. If you already received pemetrexed, the team carries forward the correct vitamin schedule.

Sources: nccp

Medicines review

Check with the team before using ibuprofen, other anti-inflammatory painkillers, aspirin or any new medicine. Kidney function and the medicine determine whether and when it needs interruption. Do not stop a prescribed medicine on your own; ask for a written plan.

Sources: smpc

Your consent conversation

Discuss whether this is maintenance or second-line treatment, the individual aim, likely benefit and uncertainty, alternatives including not proceeding, common and serious harms, possible lasting kidney or nerve effects, fertility and pregnancy, vitamin and steroid medicines, practical burdens and your right to reconsider. A discussion aid supports but does not replace informed consent.

Sources: nccp

TREATMENT REFERENCE · 03

Treatment day

Assessment comes before dose authorisation.

Your infusion

Pemetrexed is infused into a vein over about 10 minutes on Day 1. The whole visit is longer because it includes blood-result review, symptom checks, medicines, preparation and observation. The team can use a cannula or suitable existing line; this regimen has no routine take-home infusion pump.

Sources: nccp

Tell the nurse immediately

Report feeling unwell, itching, swelling, wheeze, chest tightness or pain, burning or redness around the drip.

Sources: nccp

A delay or dose change

The team may delay or reduce a dose so that blood counts or symptoms can recover. A change does not itself mean the treatment has failed; ask why it is needed and when review is planned.

Sources: nccp

TREATMENT REFERENCE · 04

Care between visits

Keep a short symptom record and ask for help early.

Vitamins and steroids

Follow the written folic acid, B12, steroid and anti-sickness plan. Tell the team promptly about missed doses or vomiting that prevents you keeping medicines down.

Sources: nccp

Mouth, bowel and skin symptoms

Report mouth ulcers, difficulty eating or drinking, persistent diarrhoea, rash and changes in daily function. Severe symptoms or sudden illness need urgent advice.

Sources: smpc · hse

Daily life and support

Balance activity with rest. Ask for help with fatigue, eating, transport, work, finances or emotional wellbeing. Discuss what matters most to you at each review.

Sources: hse

Pregnancy, fertility and feeding

Tell the team immediately if pregnancy is possible. Discuss fertility preservation and contraception before treatment and confirm the medicine-specific duration after the final dose. Breastfeeding must stop during pemetrexed.

Sources: smpc

When to get urgent help

Keep your oncology contact number available day and night.

Call immediately

Call your oncology 24-hour number immediately for a temperature of 37.5°C or above, 36°C or below, shivering or sudden illness. Call urgently for new breathlessness, chest pain, bleeding, severe or persistent diarrhoea, vomiting despite medicines, inability to drink, very little urine or a spreading or blistering rash. Do not wait for the next visit; serious infection can occur without a high temperature.

Sources: hse · smpc

When you call

Give the medicine name, last treatment date and symptoms. Say whether you can drink, pass urine and manage everyday activities. Do not delay a call to collect these details.

Sources: hse

Symptoms after stopping

New or worsening symptoms after the last dose still need advice. Explain that you recently had chemotherapy.

Sources: smpc · hse

TREATMENT REFERENCE · 06

Review and the next plan

Treatment decisions consider benefit, harms and your priorities.

Scans and disease review

Scans and other tests follow the Consultant treatment plan. There is no fixed scan interval in this regimen, and side effects do not reliably show whether treatment is working.

Sources: nccp

When treatment stops

Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

Sources: nccp

After the final pemetrexed

Continue folic acid for 21 days after the last pemetrexed. Agree follow-up for persistent blood-count, kidney, nerve or other symptoms and confirm which support medicines stop.

Sources: nccp

TREATMENT REFERENCE · 07

Vaccines and radiotherapy

Tell the team about planned vaccines and any past or planned radiotherapy.

Check before a vaccine

The team must review the vaccine type and timing. Yellow fever vaccine must not be given with pemetrexed; other live vaccines are also not usually suitable.

Sources: smpc

Past or planned radiotherapy

Tell the team, even if radiotherapy was years ago. Report new cough, breathlessness or a skin reaction in a previously treated area.

Sources: smpc

Evidence depends on the treatment setting

Maintenance and second-line studies answer different questions.

Maintenance studies found longer disease control or survival with pemetrexed than placebo in selected patients whose cancer had not progressed after initial chemotherapy. One studied starting pemetrexed after a different chemotherapy combination; another studied continuing it after pemetrexed and cisplatin. A separate second-line study found similar survival to docetaxel with fewer important blood-count side effects. These historical comparisons do not predict an individual outcome or describe pemetrexed plus immunotherapy.

Sources: hanna · jmen · paramount

PREPARE FOR THE CONVERSATION

Questions worth asking

Does having more side effects mean treatment is working?

No. Side effects do not reliably show whether the cancer is responding. Report them so they can be treated. Your team assesses benefit using the tests and review plan appropriate to your cancer.

Why might a dose be delayed or reduced?

Your team may allow recovery, investigate a symptom or reduce the chance of lasting harm. Ask which drug changes, why, and when review is planned. A delay does not itself mean treatment has failed.

What should I say when calling?

Give the regimen name, last treatment date, symptoms and when they started. Explain whether you can eat, drink, pass urine and manage usual activities. Give your temperature if available, but do not delay calling to collect details.

What belongs in the consent conversation?

The individual aim and likely benefit, uncertainty, reasonable alternatives including not proceeding, common and serious risks, lasting effects, fertility and practical concerns. You can ask for an interpreter, more time or another explanation and can reconsider your decision.

How many cycles should I use this passport for?

The passport provides reusable recording pages, not a prescribed course length. Your oncology team sets the duration and can provide extra pages.

Can I use this page to decide if blood results are safe?

No. Treatment decisions require the clinical protocol, symptoms, previous toxicity, organ function and the prescriber’s review. The patient page explains the checks and does not authorise treatment.

Is this the same as pemetrexed with pembrolizumab?

This guide covers pemetrexed on its own for maintenance or second-line treatment of selected non-squamous non-small-cell lung cancer. If pembrolizumab or another immunotherapy is also prescribed, use the matching combination regimen and its additional consent, monitoring and urgent advice. Pemetrexed with pembrolizumab maintenance is a different treatment plan.

What is maintenance treatment?

Maintenance means continuing cancer treatment after the initial chemotherapy has finished and the cancer has not progressed. In this regimen pemetrexed is the only anti-cancer medicine. Your team confirms whether it is being continued from your earlier combination or started as a new maintenance medicine.

How many cycles will I have?

Treatment is planned every three weeks while it is helping and side effects remain acceptable. There is no fixed total number of cycles in this regimen. Scans, symptoms, blood results, your priorities and the oncology review guide continuation, delays, changes or stopping.

Can I stop the vitamins when platinum treatment stops?

Do not stop them because platinum stops. Folic acid and B12 protect against pemetrexed toxicity and continue according to the pemetrexed plan. The team checks your existing vitamin schedule at a phase change.

How long will I be at the unit?

Pemetrexed is infused into a vein over about 10 minutes on Day 1. The whole visit is longer because it includes blood-result review, symptom checks, medicines, preparation and observation. The team can use a cannula or suitable existing line; this regimen has no routine take-home infusion pump.

Can I take my usual painkillers?

Check with the team before using ibuprofen, other anti-inflammatory painkillers, aspirin or any new medicine. Kidney function and the medicine determine whether and when it needs interruption. Do not stop a prescribed medicine on your own; ask for a written plan.

Can I have vaccinations while on treatment?

Check first with the oncology team. Live vaccines, including yellow fever vaccine, can be unsafe during treatment. Other vaccines may still be appropriate; do not assume every vaccine should be stopped.

> **Try explaining the plan back:** Can you explain whether this is maintenance or second-line treatment, which vitamin and steroid medicines continue, when B12 is due, and how to get urgent help?

SOURCE TRANSPARENCY

Where this information comes from

Regimen schedule: NCCP 00222 v5. Medicine explanations and safety information: sources below. The timeline is an educational illustration, not a measured risk curve or an individual prediction.

- NCCP 00222 v5 · official regimen
- EMA ALIMTA product information and patient leaflet
- HSE chemotherapy side effects and urgent advice
- NCCP NCIS Medical Oncology antiemetics V8, Table 1 low-risk row
- Hanna 2004: second-line pemetrexed versus docetaxel
- JMEN 2009: switch maintenance pemetrexed versus placebo
- PARAMOUNT final OS 2013: continuation maintenance after pemetrexed/cisplatin

SACTCheck · Developed by Dr Paul O’Brien · Sources checked 2026-11-02.
This information supports your treatment discussion. Your oncology team confirms your personal treatment and urgent-care plan.

Written source addresses

https://healthservice.hse.ie/documents/6935/222_V5_PEMEtrexed_Monotherapy_.pdf

https://www.ema.europa.eu/en/documents/product-information/alimta-epar-product-information_en.pdf

https://www2.hse.ie/conditions/chemotherapy/side-effects/

https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf

https://pubmed.ncbi.nlm.nih.gov/15117980/

https://pubmed.ncbi.nlm.nih.gov/19767093/?dopt=Abstract

https://pubmed.ncbi.nlm.nih.gov/23835707/